# Science practice booklet: Grade 4 expectations

Grade 4 · Practice booklet

Name: _____

### What this booklet is

This is general practice written for Science at Grade 4 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

### How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

## Section 1: Living things

Standards practised: 4-LS1-1, 4-LS1-2

### A worked example

**A camel has wide, flat feet. Explain how this structure helps the camel survive in the desert.**

1. Name the structure: wide, flat feet.
2. Think about the place where the camel lives: soft, loose sand.
3. Wide feet spread the camel's weight over a bigger area, so the feet do not sink deep into the sand.
4. Link it to survival: the camel can walk long distances to find food and water.

**Wide, flat feet spread the weight so the camel does not sink in soft sand, and it can travel far to find food and water.**

### Questions

1. A camel's hump stores fat. How does this help a camel survive in the desert?

2. A camel has long eyelashes and can close its nostrils. What problem do these structures help with?

3. The Arabian oryx has a pale, almost white coat. How can this help it in a hot, sunny desert?

4. Mangrove trees grow in salty, muddy water along the coast. Their roots stick up out of the mud. Give one way this structure helps the tree.

5. A falcon has a sharp, curved beak and strong claws. What are these structures used for?

6. What are two jobs of the roots of a date palm?

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7. Many desert animals come out to hunt or look for food at night. How does this help them survive?

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8. Match each sense organ to what it detects: eyes, ears, skin, tongue. Choose from: sound, light, taste, touch and heat.

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9. Hamdan sees a ball coming towards him and catches it. Put these steps in order: the brain decides what to do; the eyes detect the ball; the muscles move the arms; signals travel along nerves to the brain.

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10. Maryam touches a cup and feels that it is hot, so she puts it down. Which body parts work together to make this happen?

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11. Salem says an animal's body parts help it survive only when they work together with its senses and brain. Do you agree? Explain with one example.

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## Section 2: Earth and space

Standards practised: 4-ESS1-1, 4-ESS2-1, 4-ESS2-2, 4-ESS3-1, 4-ESS3-2

### A worked example

**Over many years, heat in the day and cool at night make a rock crack into small pieces. Then the wind carries the small pieces away. Which part is weathering and which part is erosion?**

1. Weathering is when rock is broken into smaller pieces.
2. Erosion is when the pieces are moved from one place to another by wind, water or ice.
3. Cracking into small pieces is weathering.
4. The wind carrying the pieces away is erosion.

**Cracking is weathering; the wind carrying the pieces away is erosion.**

### Questions

1. A cliff has layers of rock that have not been disturbed. Which layer is the oldest, the top layer or the bottom layer?

2. Rock layers in the Hajar mountains contain fossils of sea shells. What does this tell us about the place long ago?

3. What is the difference between weathering and erosion?

4. Is each of these weathering or erosion? (a) Water freezes in a crack and splits a rock. (b) Wind blows sand from one side of a dune to the other.

5. Sand dunes slowly change their shape and position over the years. What causes this?

6. After heavy rain, water rushes down a dry valley (a wadi) and carries away soil and stones. Is this weathering or erosion? What may happen to the land over many floods?

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7. Ghaf trees and grasses grow on the edge of some dunes. How can their roots help the land?

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8. A map of Earth has a key that shows blue for water, brown for mountains and yellow for sandy desert. Why is a key useful?

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9. The sun and the wind are renewable energy sources. Oil and natural gas are not. What does renewable mean?

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10. A solar park uses sunlight to make electricity. Give one good thing about this compared with burning oil or gas.

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11. Strong winds in a sandstorm can be harmful. Give two ways that people can stay safe.

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12. Sand is blowing onto a road and covering it. Describe a solution that people could build to reduce the problem.

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## Section 3: Matter, forces and energy

Standards practised: 4-PS3-1, 4-PS3-2, 4-PS3-3, 4-PS3-4, 4-PS4-1, 4-PS4-2, 4-PS4-3, 3-5-ETS1-1, 3-5-ETS1-3

### A worked example

**A small solar-powered torch is left in the sun for the day. At night it gives light. Describe how energy changes form in the torch.**

1. Start with the energy that comes in: light from the sun reaches the solar panel.
2. The panel changes the light energy into electrical energy, and this is stored in a battery.
3. At night the battery gives electrical energy to the bulb.
4. The bulb changes the electrical energy into light (and a little heat).

**Sunlight (light energy) becomes electrical energy in the panel, is stored in the battery, and becomes light energy again in the bulb.**

### Questions

1. Omar rolls a ball gently, then rolls it very fast into a box of cardboard tubes. Which roll hits the tubes harder? What does that tell us about the ball's energy?

2. When you rub your hands together quickly, they feel warm. What kind of energy changes into heat?

3. Name the energy change in each: (a) a solar panel; (b) an electric fan.

4. A spoon is left in a hot cup of karak. After a few minutes the handle feels warm. How did the heat get to the handle?

5. A moving ball hits a ball that is resting. The resting ball rolls away. What happened to the energy?

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6. A wind turbine in a wind farm is turned by the wind. Name the energy the wind has and the kind of energy the turbine can make.

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7. A floating ball bobs up and down on sea waves. Does the wave carry the ball all the way to the beach? Explain.

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8. In a dark room with no light you cannot see a book. When a lamp is switched on you can. Explain why.

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9. Two friends plan to send messages with a torch: two flashes mean yes and three flashes mean no. Why must they agree on the pattern first?

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10. Name one other way, besides flashing a light, that a pattern of sound or light can be used to send information.

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11. Noura is designing a shade for the school playground. It must keep the sun off the children (criteria) and cost less than AED 500 (constraint). What is the difference between a criterion and a constraint?

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12. Khalid tests two shade designs by measuring the temperature under each one. Why should he test both in the same place and at the same time of day?

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# Answer guide for parents

## Section 1: Living things

1. The camel can use the stored fat for energy when there is little food, so it can travel a long time without eating.
2. They keep blowing sand out of the eyes and nose, for example in a sandstorm.
3. A pale coat reflects a lot of sunlight, which helps the oryx stay cooler.
4. The roots help hold the tree steady in soft mud and let the tree take in air (one of these is enough).
5. Catching and holding prey, and tearing food so it can be eaten.
6. They take in water from the ground and hold the tree firmly in place.
7. The night is cooler, so they avoid the heat of the day and lose less water from their bodies.
8. Eyes: light. Ears: sound. Skin: touch and heat. Tongue: taste.
9. The eyes detect the ball; signals travel along nerves to the brain; the brain decides what to do; the muscles move the arms.
10. The skin senses the heat, nerves carry the signal to the brain, and the brain tells the muscles to move the hand away.
11. Yes. For example, a falcon's eyes spot a small animal far away, the brain processes the information, and its wings and claws then help it catch the prey.

## Section 2: Earth and space

1. The bottom layer, because it was laid down first and newer layers formed on top of it.
2. The land was once covered by the sea.
3. Weathering breaks rock into smaller pieces. Erosion moves the pieces from one place to another.
4. (a) Weathering. (b) Erosion.
5. Wind picks up sand grains, carries them and drops them in a new place, so the dune moves and changes shape.
6. Erosion. The moving water can carve the valley deeper and wider over time.
7. The roots hold the soil and sand in place, so the wind and water carry away less of it.
8. It tells us what each colour or symbol on the map means.
9. A renewable source will not run out, or it is replaced quickly by nature.
10. It uses a source that does not run out, and it does not give off smoke or the gases made by burning fuel.
11. Stay indoors, close windows and doors, and cover the nose and mouth with a mask or cloth if outside (any two).
12. For example: a fence or a row of plants beside the road to block the wind and catch the sand before it reaches the road.

## Section 3: Matter, forces and energy

1. The fast roll hits harder. A faster object has more energy of motion.
2. The energy of motion (from moving your hands).

3. (a) Light energy becomes electrical energy. (b) Electrical energy becomes energy of motion (and some sound).
4. Heat moved along the spoon from the hot karak to the cooler handle.
5. Some energy moved from the first ball to the second ball, so the second ball began to move.
6. The wind has energy of motion. The turbine makes electrical energy.
7. Not by itself. The water moves up and down as the wave passes, so the ball mostly bobs in the same place while the wave's energy travels on.
8. Light from the lamp hits the book and bounces off it into our eyes. We see an object when light from it reaches our eyes.
9. A pattern only carries information if both people know what each pattern means.
10. For example: a car horn pattern, a siren, a school bell pattern, a lighthouse flashing, or sound signals on a phone (any sensible answer).
11. A criterion is what the design must do to succeed. A constraint is a limit, such as cost, time or materials.
12. So the test is fair: only the design is different, so any change in temperature comes from the design.